Relation-aware simplicial attention (static text-relation route)

Zero-shot relational-table pretraining where **neighbor aggregation is conditioned on the foreign-key type**, and each relation type is represented by the **text embedding of its schema description** (e.g. `"director between movie and person"`).

The key empirical result on this branch: freezing that text-relation prior (`update_relations=False`) and holding the init with a long Muon warmup gives a zero-shot AUROC of **~0.87** on the held-out DB — well above the 0.816 target and the previous 0.8169 record — reached in the first ~100 steps.

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1. The idea in one line

Conventional neighbor attention asks "*how relevant is child* `s` *to parent* `q`*?*". Here the score also depends on **which relation** connects them:

```
score(q, s, r) = Σ_d  q_d · k^cell_s,d · k^rel_r,d          (per head)
value(s, r)    = v^cell_s ⊙ v^rel_r                          (FiLM-style gate)
```

`r` is a per-relation-type embedding, initialised from text and — on the winning route — **kept frozen** through all blocks.

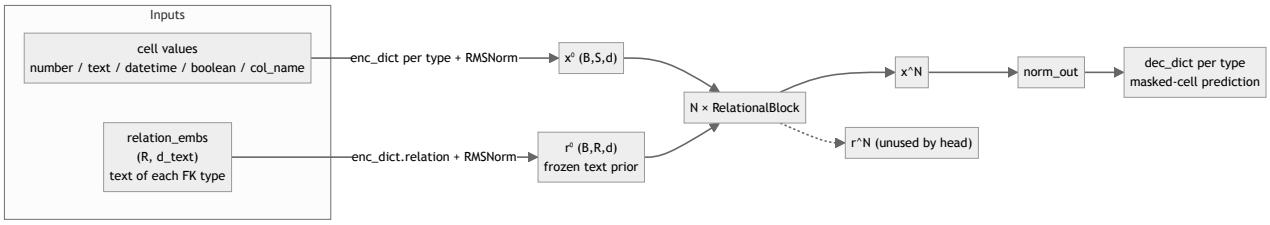
> Note (see `project_simplicial_diagnosis`): algebraically this is a rank-1 DistMult/FiLM operator, not a true 2-simplex — `r` is glued to the edge and the softmax is segmented only by `q`. The name "simplicial" is historical.

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2. Two streams

The model carries two token streams through every block:

- **`x` — cells** `(B, S, d)`: one token per (row, column) cell in the sampled subgraph.
- **`r` — relations** `(B, R, d)`: one token per foreign-key *type*, shared across all edges instantiating it.

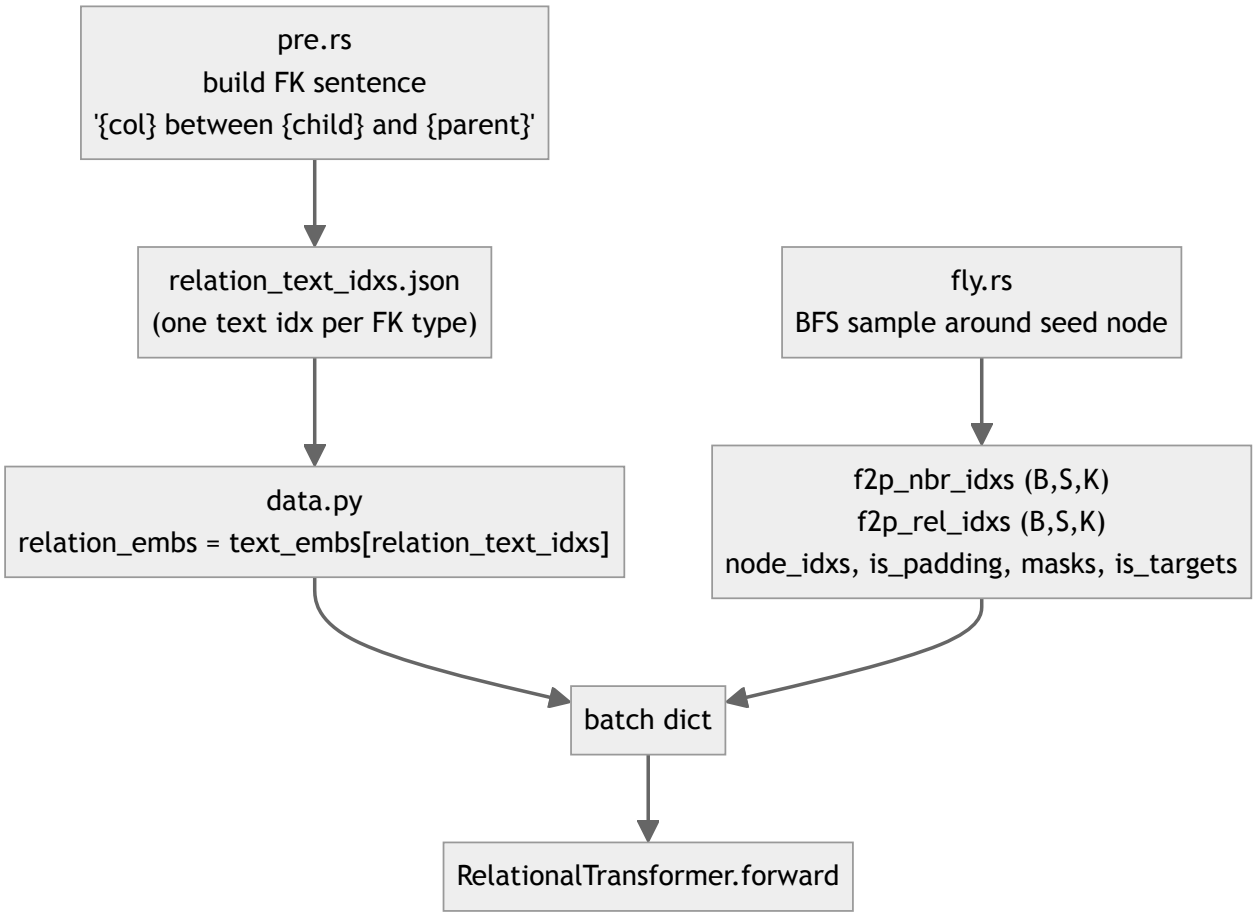


Only `x` reaches the prediction head; `r` conditions `x` but is never decoded.

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3. Data pipeline (leak-safe sampling)

The Rust extension (`rustler/`) builds the relation text and samples a temporal, leak-free subgraph per seed node.

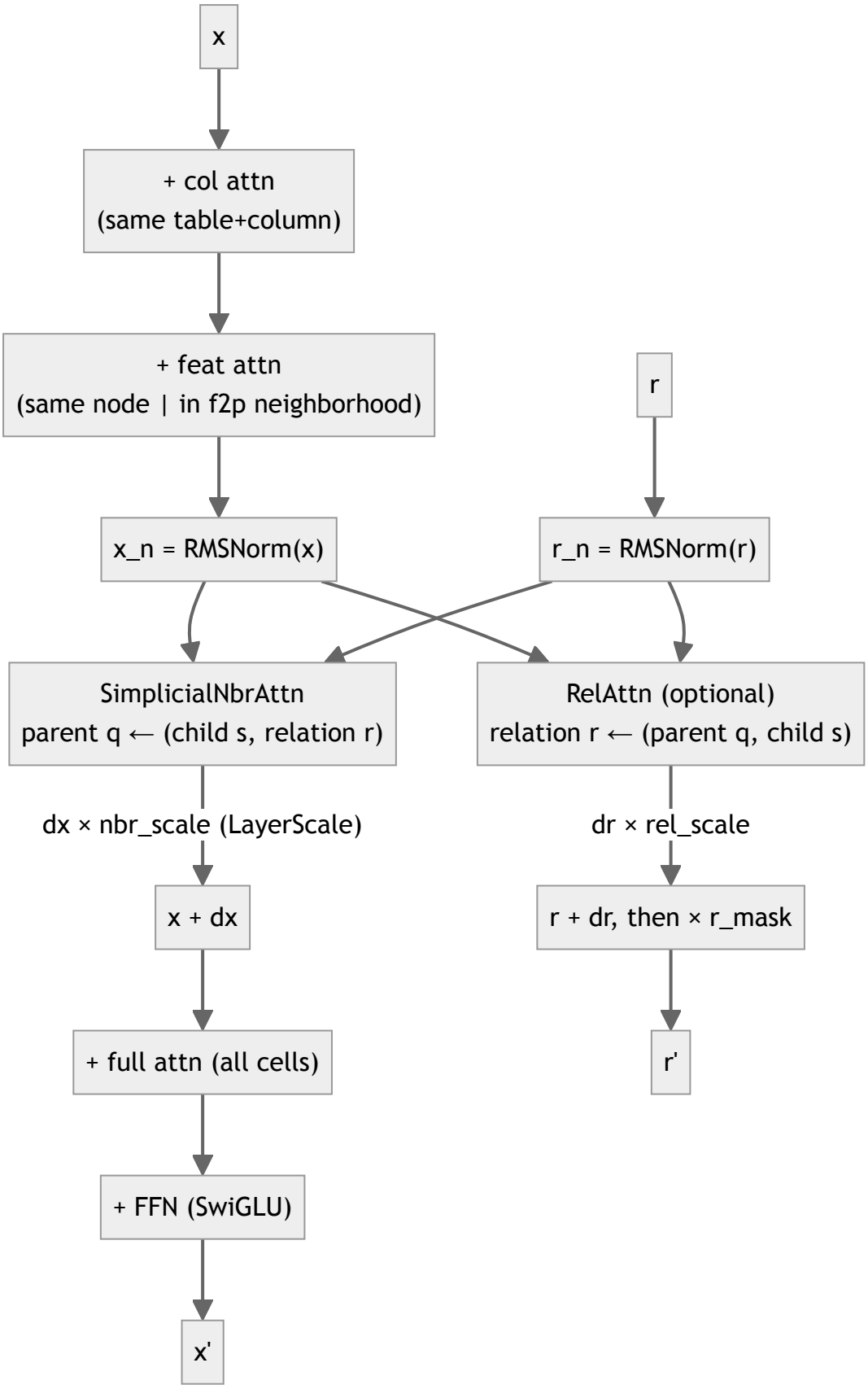


Leak prevention in `fly.rs` (validated):

vector	mechanism	ref
temporal	edges with `timestamp > seed_node.timestamp` are skipped	`fly.rs:447`
target value	target cell gets `is_target` + `masks=True`; model uses `mask_emb` on input, value only as label	`fly.rs:528–532`
leaky columns	`columns_to_drop` removed from seed **and** same-timestamp nodes	`fly.rs:482–483`
holdout	`train_tasks` excludes `holdout_db`; val loads only the holdout artifact → zero-shot	`nanort_train.py:79–82`

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4. Inside a `RelationalBlock`



- `col` / `feat` / `full` are standard masked attentions (flex_attention).
- `SimplicialNbrAttn` updates cells; `RelAttn` updates relations.

- **`update_relations=False`** drops `RelAttn` entirely → `r` stays r^0 (the frozen text prior). This is the winning configuration.
- `nbr_scale` / `rel_scale` are per-channel LayerScale gates on the deltas (`simplicial_layerscale_init`, default 1.0 = plain residual).
- The relation stream is re-masked with `r_mask` after each block (defensive; the relation encoder has a bias).

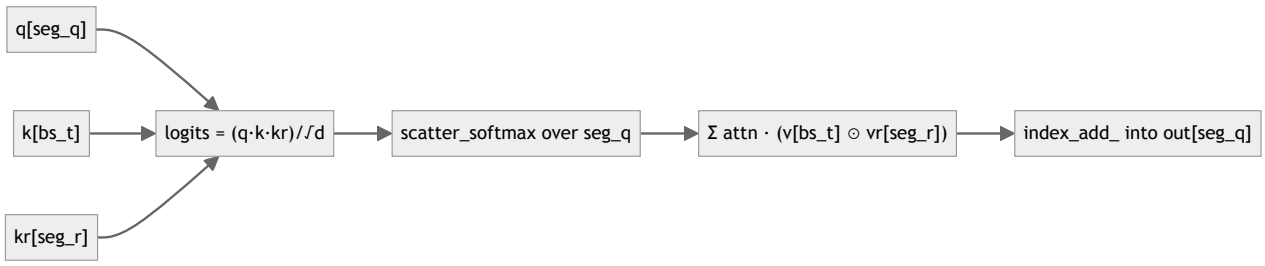
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5. Sparse edge-scatter (why it stays in memory)

Dense logits `(B, H, S, S, K)` would explode. `build_rel_batch` compiles the valid FK slots into flat edge triples once per batch:

```
seg_q = b·S + q   (parent cell)     → gathers queries, segments softmax, scatters cel
bs_t  = b·S + s   (child cell)      → gathers keys/values
seg_r = b·R + r   (relation type)   → gathers relation K/V, segments RelAttn
```

`SimplicialNbrAttn` then does, over the `T` edges (all `index_select` / `index_add_`, so index traffic stays 1-D):



Per-head RMSNorm (`qk_norm`) is applied to `q`, `k`, `kr` before the triple product to keep the trilinear logit scale from collapsing the softmax entropy.

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6. SIGReg (optional geometry regularization)

`SIGReg` regularizes the per-cell hidden state after block `sigreg_layer` (row/col views derived from `node_idxs` and the (table,col) key). Training-only auxiliary loss: `(1−λ_row−λ_col)·L_sup + λ_row·row + λ_col·col`, trained by a separate fp32 AdamW. Disabled by default (`sigreg_layer=None`).

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7. Training recipe (the result)

ingredient	value	why
relation stream	**static** (`update_relations=False`)	the dynamic `RelAttn` corrupted the text prior (one-hot, `r` norm ×4)
simplicial	all blocks	conditions neighbor aggregation on the clean text prior
optimizer	Muon (2-D weights) + AdamW (rest)	
muon warmup	**3000 steps**	holds the strong init; the model drifts off it even at tiny LR
checkpoint	best-by-val-AUROC	AUROC peaks early (~step 25) and decays, so the final step is not the best

Metric: `time_to_target` = wall-clock seconds until `val/auroc_zero_shot` first crosses **0.81621** on the held-out DB.

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8. Diagnosis summary

- The text-relation prior alone carries strong zero-shot signal; the masked-cell reconstruction objective *degrades* it as training proceeds.
- Low LR / long warmup **slow** the decay; whether SIGReg **stops** it is under test.
- The current simplicial operator is FiLM rank-1 (redundant with the bilinear `feat` path); a genuine order-2 operator (free second endpoint + joint softmax) is the open research direction.